

| Outlook  | Temp (°F) | Humidity (%) | Windy? | Class             |
|----------|-----------|--------------|--------|-------------------|
| sunny    | 75        | 70           | true   | <i>play</i>       |
| sunny    | 80        | 90           | true   | <i>don't play</i> |
| sunny    | 85        | 85           | false  | <i>don't play</i> |
| sunny    | 72        | 95           | false  | <i>don't play</i> |
| sunny    | 69        | 70           | false  | <i>play</i>       |
| overcast | 72        | 90           | true   | <i>play</i>       |
| overcast | 83        | 78           | false  | <i>play</i>       |
| overcast | 64        | 65           | true   | <i>play</i>       |
| overcast | 81        | 75           | false  | <i>play</i>       |
| rain     | 71        | 80           | true   | <i>don't play</i> |
| rain     | 65        | 70           | true   | <i>don't play</i> |
| rain     | 75        | 80           | false  | <i>play</i>       |
| rain     | 68        | 80           | false  | <i>play</i>       |
| rain     | 70        | 96           | false  | <i>play</i>       |

## Decision Tree

- Recall cricket-ball vs tennis-ball example
- $x = (x_1, x_2) \in R^2$
- $y \in \{0,1\}$  1 is 'cricket' 0 is 'tennis'
- If weight of a ball is  $> 100$  gm, ball is cricket ball else tennis ball
- Mathematically,
- $f(x) = 1$  if  $x_1 > 100$   
 $= 0$  otherwise

# ID3

- Iterative Dichotomizer Ross Quinlan in 1980s
- Which feature should be used to make decision?
  - The one which will separate all classes well
  - Information theoretically, the one from which we have highest information gain
  - How to measure Information Gain?

# Entropy

- I need to send you one of the two observations
  - It is raining today
  - It is not raining today
- How many bits I need to send you?
- One bit
  - = '0' means It is raining today
  - = '1' means It is not raining today
- Suppose there is four messages: Two bits

- It is raining today and is sunny
  - It is raining today and is cloudy
  - It is not raining today and is sunny
  - It is not raining today and is cloudy
- 
- Suppose it is never going to be rainy and sunny
  - Suppose I am in desert it is never going to rain and is always day is sunny
    - Do I need to really send you the bit?

- Can we do better than 'two bits' in the three message or do we need one bit '0' to say no rain everyday?
- Every system of discrete signals has some 'information' and we need those many bits to represent the system
- Entropy =  $\sum_i p_i * \log\left(\frac{1}{p_i}\right)$

# Information Gain

- Entropy  $H(S) = \sum_{i \in [K]} \frac{S_i}{S} * \log \frac{S}{S_i}$
- Information Gain due to  $A$ 
  - $IG(S, A) = H(S) - H(S|A)$   
 $= H(S) - \sum_t \frac{S_t}{S} H(S_t)$   
where,  $t \in T$  is the different values present in  $S$  for  $A$

Detail

Compact

Column

5 of 5 columns

| ▲ Outlook |     | ▲ Temperature |     | ▲ Humidity         |  | ▲ Wind |     | ✓ Play Tennis                       |  |
|-----------|-----|---------------|-----|--------------------|--|--------|-----|-------------------------------------|--|
| Sunny     | 36% | Mild          | 43% | 2<br>unique values |  | Weak   | 57% | <br>true<br>9 64%<br>false<br>5 36% |  |
| Rain      | 36% | Hot           | 29% |                    |  | Strong | 43% |                                     |  |
| Other (4) | 29% | Other (4)     | 29% |                    |  |        |     |                                     |  |
| Sunny     |     | Hot           |     | High               |  | Weak   |     | No                                  |  |
| Sunny     |     | Hot           |     | High               |  | Strong |     | No                                  |  |
| Overcast  |     | Hot           |     | High               |  | Weak   |     | Yes                                 |  |
| Rain      |     | Mild          |     | High               |  | Weak   |     | Yes                                 |  |
| Rain      |     | Cool          |     | Normal             |  | Weak   |     | Yes                                 |  |
| Rain      |     | Cool          |     | Normal             |  | Strong |     | No                                  |  |
| Overcast  |     | Cool          |     | Normal             |  | Strong |     | Yes                                 |  |
| Sunny     |     | Mild          |     | High               |  | Weak   |     | No                                  |  |
| Sunny     |     | Cool          |     | Normal             |  | Weak   |     | Yes                                 |  |
| Rain      |     | Mild          |     | Normal             |  | Weak   |     | Yes                                 |  |
| Sunny     |     | Mild          |     | Normal             |  | Strong |     | Yes                                 |  |
| Overcast  |     | Mild          |     | High               |  | Strong |     | Yes                                 |  |
| Overcast  |     | Hot           |     | Normal             |  | Weak   |     | Yes                                 |  |
| Rain      |     | Mild          |     | High               |  | Strong |     | No                                  |  |